

E7 Head-to-head vs. ProvSQL — exact probabilistic query evaluation

all 3/3 instances: SPARQL_circ probability == ProvSQL, exact to float precision (max $|\Delta p| = 2e-16$). The difference is deployability:

	ProvSQL	SPARQL_circ
Exact answer probability (PQE)	✓	✓
Probability == the other system (measured)	✓	✓
Knowledge compilation (circuit → d-DNNF)	✓	✓
Non-monotone OPTIONAL / MINUS	✓	✓
Native RDF / SPARQL (no relational remodelling)	✗	✓
Runs on an UNMODIFIED engine	✗	✓

Timings (both sub-10 ms on these instances) are NOT a fair speed comparison — ProvSQL builds+compiles in-DB, our client compiles the returned circuit — so E7 is framed as parity + deployability, per the pre-registration.